

Chaos

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This exercise demonstrates synchronization in a model based on Lorenz dynamics, corresponding to the mechanical system known as the **Lorenz Waterwheel**.

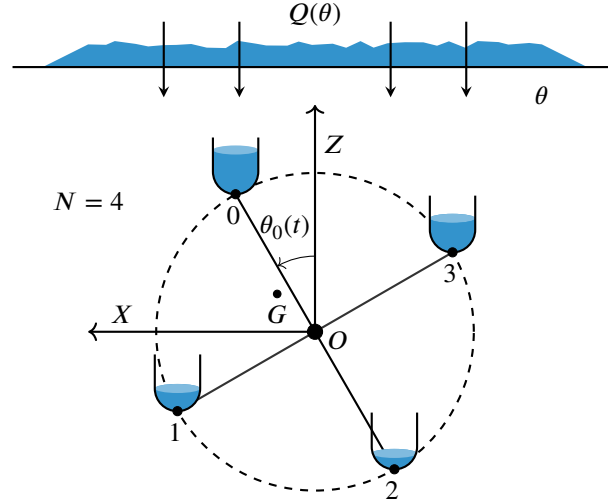


Figure 1: Malkus waterwheel model

Synchronization of two chaotic systems

Although the synchronization of periodic systems (e.g., pendulums) has been studied for centuries, the discovery that chaotic systems can synchronize is quite recent ([Fujisaka 1983 Japan](#), [Afraimovich 1986 USSR](#), [Pecora 1990 USA](#)). This was a surprise because a perturbation in a chaotic system tends to grow exponentially. However, synchronization is possible if the systems are similar and are adequately coupled. Under these conditions, regardless of differing initial conditions, the two coupled systems will converge toward an identical evolution.

Consider **System 1**, described by 3 variables $M(t) = (X(t), Y(t), Z(t)) \in \mathbb{R}^3$ subject to the following law of motion:

$$\frac{dX}{dt} = \sigma(-X + Y), \quad \frac{dY}{dt} = rX - Y - XZ, \quad \frac{dZ}{dt} = XY - Z \quad (1)$$

where $\sigma, r > 0$ are constants. Now, consider **System 2**, described by 3 variables $m(t) = (x(t), y(t), z(t)) \in \mathbb{R}^3$ subject to the law of motion:

$$\frac{dx}{dt} = \sigma(-x + y) - K(x - X), \quad \frac{dy}{dt} = rx - y - xz, \quad \frac{dz}{dt} = xy - z, \quad (2)$$

where $K > 0$ is a coupling constant and σ, r are identical to those in (0.1). Note the influence of System 1 on System 2 through the presence of the variable $X(t)$.

Let $\delta(t) = m(t) - M(t) \in \mathbb{R}^3$ represent the error vector between the states of the two systems. Note that “synchronization” means $\|\delta(t)\| \rightarrow 0$ as $t \rightarrow +\infty$.

(a) Write the equations of motion for the error $\delta(t) = (\delta_x(t), \delta_y(t), \delta_z(t)) \in \mathbb{R}^3$ in the form $\frac{d\delta}{dt} = F(\delta, M)$, which must be explicitly detailed.

(b) Let $V(t) := \|\delta(t)\|^2$. Express $\frac{dV}{dt}$ in the form:

$$\frac{dV}{dt} = -2 \left[(\delta_z + \alpha\delta_x)^2 + (\delta_y + \beta\delta_x)^2 + \gamma\delta_x^2 \right]$$

specifying the expressions for α, β, γ .

(c) It is given that $D := \{M = (X, Y, Z) \in \mathbb{R}^3, X^2 < 4\sigma r, Y^2 \leq r^2, 0 \leq Z \leq 2r\}$ is a trapping region for System 1, meaning that if the initial condition $M(0) \in D$, then $M(t) \in D$ for all $t \geq 0$. Show that there exists a threshold coupling constant $K_0 > 0$, which must be specified, such that for any initial condition $M(0) \in D$ and constant $K > K_0$, then $\frac{dV}{dt} < 0$. Conclude.

References

- [1] Hirokazu Fujisaka and Tomoji Yamada. Stability Theory of Synchronized Motion in Coupled-Oscillator Systems. *Progress of Theoretical Physics*, 69(1):32–47, 1 1983.
- [2] V. I. Arnold, V. S. Afraimovich, Yu. S. Il'yashchenko, and L. P. Shil'nikov. *Modern Problems in Mathematics. The Fundamental Trends*. VINITI Akad. Nauk USSR Moscow, 1986.
- [3] Louis M. Pecora and Thomas L. Carroll. Synchronization in Chaotic Systems. *Physical Review Letters*, 64(8):821, 1990.